Corporate credit risk assessment

Portfolio model review | Independent work sample

Audience: technical reviewers and credit analysts. All borrowers and financial observations are synthetic.

Assessment

The logistic model ranks defaults more effectively than the benchmark tree in the held-out cohort: ROC-AUC is 0.745 versus 0.711. Its mean predicted default probability is 21.9%, compared with an observed 12-month default rate of 22.8%. The transparent logistic specification remains the demonstration model for scenario analysis.

The exercise demonstrates a reproducible analytical workflow using synthetic data. The generator uses a logistic relationship, which favors the fitted logistic model. The results cannot establish production model validity or the credit quality of any real institution or borrower.

Population and validation design

Cohort	Snapshot years	Borrowers	Default rate
Training	2017 to 2019	2,400	20.4%
Validation	2021	800	20.9%
Held-out test	2023 to 2024	1,598	22.8%
Maturity embargo	2020 and 2022	1,600	Excluded

Each borrower appears once. The target records default during the next 12 months, with test outcomes matured by the end of 2025. Maturity embargoes exclude 2020 and 2022 snapshots so training and validation outcomes finish before the next evaluation period. The workflow joins financial, repayment and macroeconomic records using explicit keys and records reconciliation exceptions.

Held out model comparison

Measure	Logistic	Decision tree
ROC-AUC	0.745	0.711
Average precision	0.454	0.385
Brier score	0.1539	0.1594
Classification threshold	0.250	0.210

Sources: data/processed/model_data.csv; outputs/model_metrics.csv; outputs/test_predictions.csv. Lower Brier scores indicate less squared probability error.

Statistical validation

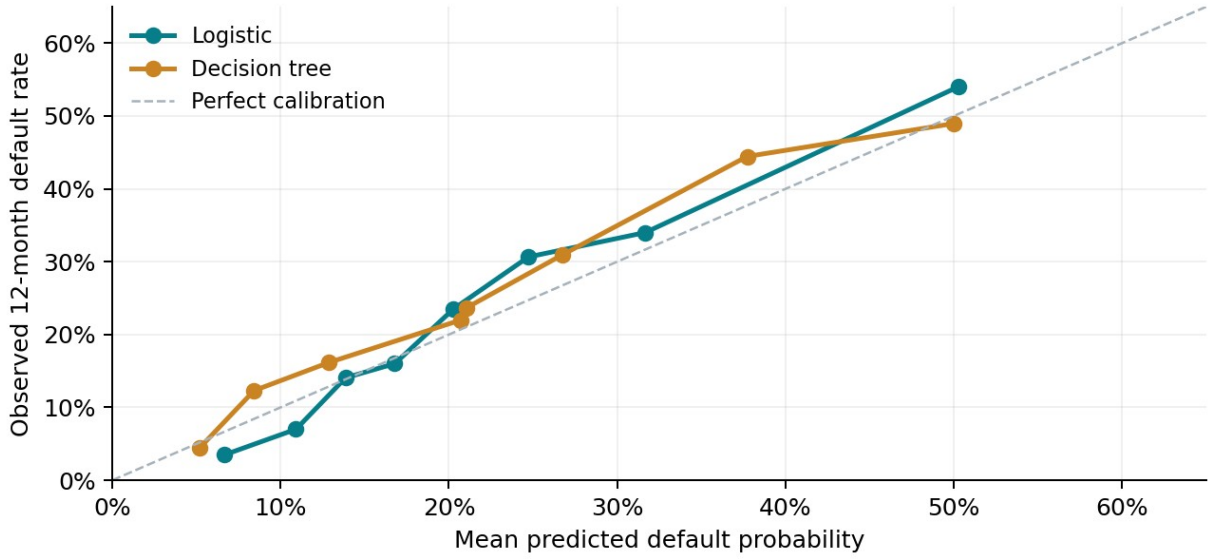
The logistic specification uses debt-to-assets leverage, debt-service coverage, borrowing interest rate, GDP growth and a prior 30-day delinquency indicator. Maximum-likelihood estimation fits the training sample only. HC1 standard errors describe the fitted synthetic sample. They do not establish causal effects or account for all dependence from shared macroeconomic conditions.

Coefficient diagnostics

Predictor	Coefficient	HC1 SE	p value
Intercept	-2.316	0.413	<0.001
Leverage	2.800	0.344	<0.001
Debt-service coverage	-0.517	0.072	<0.001
Borrowing rate	0.167	0.088	0.058
GDP growth	-0.072	0.053	0.174
Prior 30-day delinquency	1.496	0.192	<0.001

Source: outputs/coefficients.csv. Rates enter in percentage points. Leverage is debt/assets; DSCR is EBITDA/debt service. Coefficients operate on log-odds.

Probability calibration



The overall prediction gap is -0.94 percentage points. Logistic probability groups contain 199 to 200 borrowers. The highest group averages 50.3% predicted default versus 54.0% observed. Maximum training VIF is 1.42; the borrowing-rate and GDP coefficients have HC1 p values above 0.05.

Sources: outputs/calibration_bins.csv; outputs/coefficients.csv; outputs/metrics.json. Thresholds and tree settings use validation data only. HC1 inference assumes independent borrower observations.

Conditional stress review

The scenarios rescore the same held-out borrowers with frozen logistic coefficients. They change observable features directly and quantify conditional model sensitivity. They do not forecast macroeconomic transmission or simulate accounting cash flows.

Driver	Base	Adverse	Severe
Leverage	Unchanged	+0.05	+0.10
Debt-service coverage	Unchanged	Multiply by 0.85	Multiply by 0.70
Borrowing rate	Unchanged	+1 pp	+2 pp
GDP growth	Unchanged	−2 pp	−4 pp

Illustrative analyst assumptions. Leverage is bounded to [0,1]. Prior delinquency is unchanged. pp denotes percentage points.

Scenario	Mean PD	Loss proxy CNY million	Change vs base
Base	21.9%	9,782.1	—
Adverse	32.0%	13,969.1	+42.8%
Severe	44.2%	18,811.2	+92.3%

Under the adverse inputs, the expected-loss proxy increases from CNY 9,782.1 million to CNY 13,969.1 million. The severe inputs produce CNY 18,811.2 million. These comparisons hold debt balances and the 45% loss-given-default assumption constant.

The proxy sums borrower debt multiplied by modeled 12-month default probability and assumed loss given default. Debt substitutes for exposure at default. The calculation excludes undrawn commitments, collateral recoveries, discounting, staging and lifetime losses. It is not an IFRS 9 or regulatory capital calculation.

Review implications

GDP growth lies outside the training range for 800 of 1,598 base-case borrowers and every borrower under the adverse and severe scenarios. These predictions therefore involve extrapolation. Before operational use, real data would require independent outcome verification, segment validation, recalibration and a separately justified exposure and recovery model.

Reproduction and scope

Run python -m src.pipeline from the repository root to rebuild the analytical outputs. Run python reports/build_memo.py to regenerate this memo. The Excel file exposes scoring formulas and scenario controls. The repository documents execution status for optional R and SAS implementations.

Sources: outputs/scenario_summary.csv; outputs/scenario_support.csv; outputs/coefficients.csv. All figures refer to the fixed held-out cohort. Rounding may affect displayed sums.